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***B.Tech. Degree V Semester Supplementary Examination
April 2019***

**ME 15-1504 THERMAL ENGINEERING
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define adiabatic flame temperature. Explain its significance.
 (b) Explain gravimetric analysis.
 (c) Derive an expression for air standard efficiency of Diesel cycle.
 (d) Explain scavenging in IC engines.
 (e) Explain the valve timing diagram of a four stroke SI engine.
 (f) List out various methods for finding frictional power of an engine.
 (g) Explain evaporative cooling system in IC engines.
 (h) Explain throttle body injection system.
 (i) How are SI and CI engine fuels rated?
 (j) What is HUCR?



PART B

(4 × 10 = 40)

- II. Liquid octane at 25°C, 1 atm enters a well-insulated reactor and reacts with air entering at the same temperature and pressure. For steady-state operation and negligible effects of kinetic and potential energy, determine the temperature of the combustion products for complete combustion with
 (i) the theoretical amount of air,
 (ii) 400% theoretical air. (10)

OR

- III. A natural gas has the following molar analysis: CH₄, 80.62%; C₂H₆, 5.41%; C₃H₈, 1.87%; C₄H₁₀, 1.60%; N₂, 10.50%. The gas is burned with dry air, giving products having a molar analysis on a dry basis: CO₂, 7.8%; CO, 0.2%; O₂, 7%; N₂, 85%. (10)
 (i) Determine the air-fuel ratio on a molar basis.
 (ii) Assuming ideal gas behavior for the fuel mixture, determine the amount of products in kmol that would be formed from 100 m³ of fuel mixture at 300 K and 1 bar.
 (iii) Determine the percent of theoretical air.

- IV. In an engine working on Diesel cycle the weight of air and fuel supplied is 50:1. The temperature of air at the beginning of compression is 60°C and the compression ratio is 14:1. What is the ideal efficiency of the engine. CV of fuel is 42 MJ/kg. (10)

OR

(P.T.O.)

- V. Explain the effect of dissociation on temperature and brake power. How does the presence of CO affect dissociation? (10)
- VI. Explain the working of : (10)
(i) Wankel engine, and
(ii) Stratified charged engine
- OR**
- VII. (a) What are the different lubrication systems used in IC engines? (7)
(b) Discuss the merits of super charging in IC engines. (3)
- VIII. (a) Explain the combustion process in SI engines (6)
(b) Explain the factors affecting flame speed in SI engines. (4)
- OR**
- IX. Explain with figures various types of combustion chambers used in CI engine. (10)
